

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE)

April/May 2012

CE301 Parallel Computing

Date: 01.05.2012, Tuesday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

- Q-1 Answer the following questions. [11]**
- [A] Define Parallel Processing. [2]
 - [B] What is Temporal parallelism? What are the extensions of it? [2]
 - [C] Write the architectural differences between UMA and NUMA architecture. [2]
 - [D] What are the attributes of thread? [2]
 - [E] Why do we need high speed computing? [3]

- Q-2 Answer the following questions. [12]**
- [A] Briefly compare: Shared memory programming and distributed memory model. [4]
 - [B] Write a thread program for addition of two 4 X 4 matrices and store result into third matrix. [4]
 - [C] What are the problem areas of RPC? [4]

OR

- [A] What is shared memory programming model? What are the advantages of it? [4]
- [B] Differentiate between data parallelism and temporal parallelism. [4]
- [C] Explain the architecture of PVM and explain terminology used in PVM. [4]

- Q-3 Answer the following questions.(Attempt any Two) [12]**
- [A] What are the advantages and disadvantages of Data and Temporal Parallelism?
 - [B] Write down the steps to create Server service in DCE application. Explain each step in brief.
 - [C] Derive speed up equation for data parallelism.

SECTION-II

- Q-4 Answer the following questions.** [11]
- [A] What are the criteria to for selecting either loop splitting or self-scheduling method of parallelism? [2]
 - [B] Define the following terms [2]
 - (i) Starvation
 - (ii) Lightweight process
 - [C] Give some applications where high speed computing is needed. [2]
 - [D] Explain how joining of threads differs from joining of processes. [2]
 - [E] What are the disadvantages of RPC? [3]
- Q-5 Answer the following questions.** [12]
- [A] Write a short note on Flynn's classification of computer architecture. [4]
 - [B] Explain a spawning of children and group communication for message passing model? [4]
 - [C] What is condition variable in thread? Write the advantages of it. [4]
- OR**
- [A] By taking suitable example, explain what is differences between pthread_cond_broadcast() and pthread_cond_signal(). [6]
 - [B] Write a parallel program in 'C' to create histogram from an array using self-scheduling. [6]
- Q-6 Answer the following questions.(Attempt any Two)** [12]
- [A] Write a thread program to calculate standard deviation using loop splitting.
 - [B] Write a program to calculate a histogram from the values stored in any array using loop splitting.
 - [C] Write a short note on: Block Scheduling.